Doob's Proof of MLE Consistency

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Abstract

We present a concise exposition of Doob's proof of MLE strong consistency using surprisal and relative information. We clarify historical entropy terminology and explicitly verify all assumptions for the multivariate normal distribution using elementary matrix algebra.

Keywords: maximum likelihood estimator, true parameter, surprisal, relative information

1 Introduction

The consistency of the Fisher (1922) maximum likelihood estimator (MLE) is a basic result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via standard techniques dating back to Cramér (1946) and Wald (1949). A modern treatment is van der Vaart (1998). Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the Doob (1934) strong consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with no extra effort only adds to its appeal.

The contribution of this note is expository: it gives a formulation of Doob's consistency argument in surprisal/information terminology, emphasizing the role of properness and separability, and works out the multivariate normal example in detail. The proofs are designed to be self-contained throughout: in particular, the consistency argument and the matrix lemmas are developed directly.

We start with the setup and assumptions, then state and prove consistency. After that, we give a historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

2 Surprisal and Information

A random variable X is sampled repeatedly, resulting in independent and identically distributed copies $X_1, X_2, \dots$ of X. We assume the density $p(X, \theta^*)$ of X is one of a parametric family $p(X, \theta)$ of positive functions, and we wish to recover θ^* from the samples X_1, X_2, $\dots$.

Stated in this manner, one may interpret the family $p(X, \theta)$ as a probe of the underlying expectation E_*. While $p(X, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over E_*. Given this, how do we use the samples X_1, X_2, $\dots$ to recover θ^*?

Define the *surprisal* Samson (1951), Tribus (1961)

$$S(X, \theta) = \log(1/p(X, \theta)).$$

This terminology is intuitive: low probability equals high surprisal. Assume $S(X, \theta)$ is integrable for all θ. Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(X, \theta))$ a *true parameter*, and the corresponding density $p(X, \theta^*)$ a *true density*.

Equivalently, given θ^*, the *relative information* is

$$I(\theta^*, \theta) = E_*(S(X, \theta)) - E_*(S(X, \theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ. The relative information I appears in many fields, in different forms. Because of widely differing terminology, in §4 we give an account of its history.

While $p(X, \theta)$ have so far been arbitrary positive functions, we now specialize to the case where $p(X, \theta)$ are densities in the usual sense. We say θ^* is a *reference parameter* if

$$E_* \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) = 1 \qquad \text{for all } \theta. \tag{2}$$

Then the corresponding density $p(X, \theta^*)$ is a *reference density*.

Let θ^* be a reference parameter, and define E and E_θ by

$$E_*(f(X)) = E(p(X, \theta^*) f(X)) \qquad \text{and} \qquad E_\theta(f(X)) = E(p(X, \theta) f(X)). \tag{3}$$

Then the relative information takes the familiar form Donsker & Varadhan (1975), Dembo & Zeitouni (1998) (see §4 for terminology)

$$I(\theta^*, \theta) = E \left(p(X, \theta^*) \log \left(\frac{p(X, \theta^*)}{p(X, \theta)} \right) \right), \tag{4}$$

and $E_\theta(1) = 1$ for all θ. Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*, \theta) = E_* \left(j \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) \right) \geq j \left(E_* \left(\frac{p(X, \theta)}{p(X, \theta^*)} \right) \right) = j(E_\theta(1)) = j(1) = 0, \qquad \text{for all } \theta.$$

It follows that a reference density is a true density.

If θ is any other true parameter, then $I(\theta^*, \theta) = 0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(X, \theta) = t \cdot p(X, \theta^*)$. Since θ^* is a reference parameter, $t = 1$, hence $p(X, \theta) = p(X, \theta^*)$: every true density equals the reference density.

It follows there is at most one reference density; when there is a reference density, it is the unique true density. We say parameters θ, θ' *share a density* if $p(X, \theta) = p(X, \theta')$. When parameters do not share densities, there is at most one reference parameter; when there is a reference parameter, it is the unique true parameter.

The simplest example is as follows. A vector $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is a probability vector if its components are nonnegative and sum to one. Suppose $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$ is a probability vector and a random variable X satisfies $P_*(X = i) = \theta_i^*$, $i = 1, 2, \dots, d$. Then the mass function of X is $p(X, \theta^*) = \theta_X^*$. If θ is any other probability vector, the surprisal is $S(X, \theta) = \log(1/\theta_X)$, and

$$I(\theta^*, \theta) = \sum_{i=1}^{d} \theta_i^* \log(\theta_i^*/\theta_i). \tag{5}$$

Turning back to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^{n} p(X_k, \theta), \qquad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^{n} S(X_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes $E_*(S(X, \theta))$. By the strong law of large numbers (LLN), $S_n(\theta)$ converges to $E_*(S(X, \theta))$ as $n \to \infty$, with probability one, for each θ. Hence it is natural to estimate θ^* by a minimizer of $S_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ. In what follows, only E_*, $p(X, \theta)$, and true parameters enter; E, E_θ, and reference parameters play no role.

3 Consistency

In addition to integrability of $S(X, \theta)$, we make three assumptions. Our first assumption is

A (Identifiability)**.** *There is at most one true parameter.*

For example, A follows if the expected surprisal is strictly convex. Also, as we saw above, A follows if there is a reference parameter and parameters do not share densities.

A sequence of parameters *subconverges* to θ if a subsequence converges to θ. If a sequence subconverges to θ, θ is a *limit point of the sequence.* A subset K of the parameter space is *compact* if every sequence in K has limit points in K.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c. Since a proper continuous function has a minimizer and since $\hat{\theta}_n$ should minimize $S_n(\theta)$, this leads us to the second assumption

B (Properness)**.** *There is an* $N \geq 1$*, depending on the samples, and a proper* $F(\theta)$*, not depending on the samples, such that, with probability one,*

$$F(\theta) \leq S_n(\theta), \qquad \text{for all } \theta \text{ and } n \geq N. \tag{6}$$

When the parameter space is compact, every continuous function $F(\theta)$ is proper, and the proof below shows assumption B is superfluous. We will need the LLN convergence $S_n(\theta) \to E_*(S(X,\theta))$ across θ, in the sense

$$S_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{7}$$

with probability one. For this we need the third assumption

C (Continuity)**.** *The parameter space is separable and there is a continuous* $\delta(\theta,\theta')$ *and an integrable nonnegative* $g(X)$ *satisfying*

$$|S(X,\theta) - S(X,\theta')| \leq \delta(\theta,\theta')g(X) \qquad \text{and} \qquad \delta(\theta,\theta) = 0. \tag{8}$$

Under C, a standard separability argument, given in the appendix, establishes (7) with probability one. Under C, the expected surprisal $E_*(S(X,\theta))$ is continuous in θ. If moreover B holds, then by sending $n \to \infty$ in (6), $F(\theta) \leq E_*(S(X,\theta))$ for all θ, hence the expected surprisal is proper. Under A, B, and C, there is a unique true parameter. Similarly, under B and C, the mean surprisal is continuous in θ and proper, hence an MLE $\hat{\theta}_n$ exists for $n \geq N$, with probability one.

These assumptions are particularly tractable for *exponential families* Cover & Thomas (2006), Murphy (2022), Schervish (1995), van der Vaart (1998), the densities $p(X,\theta)$ where $S(X,\theta)$ is a sum of products of the form $f(\theta)g(X)$.

In (7), there is no presumption of uniform convergence. This is a key feature of Doob's proof: no uniform convergence of the mean surprisal is required. The argument instead combines properness, the LLN, and the continuity estimate in (8) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let* X_1*,* X_2*, ... be independent and identically distributed, and let* $p(X,\theta)$ *be a parametric family of positive functions. Under the above assumptions,*

1. *the expected surprisal is proper and there is a unique true parameter* θ^**,*
2. *with probability one, an MLE sequence* $\hat{\theta}_n$ *exists for sufficiently large* n*,*
3. *with probability one, any MLE sequence converges to the true parameter,* $\hat{\theta}_n \to \theta^*$*.*

Proof. The first two items were derived above. For the third item, by the LLN,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{9}$$

with probability one. Fix samples for which (6), (7) and (9) hold. We show $\hat{\theta}_n \to \theta^*$ for these samples. By (6),

$$F(\hat{\theta}_n) \le S_n(\hat{\theta}_n) \le S_n(\theta^*), \qquad \text{for } n \ge N.$$

Since $S_n(\theta^*) \to E_*(S(X, \theta^*))$, the right side is bounded. Since $F(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat{\theta}_n$. Then $\hat{\theta}_n$ subconverges to θ. By (8),

$$S_n(\theta) - S_n(\theta^*) \le S_n(\theta) - S_n(\hat{\theta}_n) \le \delta(\theta, \hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(X_k). \tag{10}$$

Since the left side of (9) is bounded as $n \to \infty$, and $\delta(\theta, \hat{\theta}_n)$ subconverges to zero, the right side of (10) subconverges to zero. By (1) and (7), the left side of (10) converges to $I(\theta^*, \theta)$, hence $I(\theta^*, \theta) \le 0$. But $I(\theta^*, \theta) \ge 0$, hence $I(\theta^*, \theta) = 0$, showing $\theta = \theta^*$. Thus θ^* is the only limit point, hence $\hat{\theta}_n \to \theta^*$ as $n \to \infty$. $\square$

4 Terminology History

The relative information $I(\theta^*, \theta)$ is often called "relative entropy" Cover & Thomas (2006), Murphy (2022). The term *entropy*, originating in thermodynamics, was coined by Clausius (1865). Shortly afterward, Boltzmann (1872) derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs (1902), building on the combinatorial interpretation of entropy Boltzmann (1877) and Planck (1901), reversed Boltzmann's sign convention, writing entropy with the sign convention used today.

Shannon (1948) independently defined the *entropy*

$$H(p) = -\sum_{i=1}^{d} p_i \log p_i.$$

von Neumann (1932) had earlier defined quantum entropy. According to a widely repeated account attributed to Shannon and reported in Tribus & McIrvine (1971), von Neumann suggested the name "entropy" when Shannon was casting about for terminology after developing his communication theory in the mid-1940s; later Shannon (1982) expressed doubt that the conversation took place. Whatever the origin of the name, in his 1948 paper, Shannon explicitly links his H to Boltzmann's: "H is then, for example, the H in Boltzmann's H theorem," connecting his terminology to the earlier usage despite the difference in sign conventions.

The relative information $I(\theta^*, \theta)$ (4) is jointly convex in the densities $p(X, \theta^*)$ and $p(X, \theta)$, as is apparent in the special case (5), whereas the entropy $H(p)$ above is strictly concave in p. From this perspective, the terminology "relative entropy" can seem counterintuitive. This counter-intuition extends into the Python `scipy.stats` package: `entropy(p)` returns the concave H, whereas `entropy(p,q)` returns the convex I. The issue is the conceptual opposition between entropy and information: entropy is concave, whereas information is convex.

Indeed, based on curvature, one might argue that *relative entropy* should be $-I(\theta^*, \theta)$. Similar remarks apply to the "cross-entropy" loss J of machine learning. Since $J = I$ when the class label p is $0, 1$, or J differs from I by a constant, when $0 < p < 1$, one might analogously call J the *cross-information*, and $-J$ the *cross-entropy*.

While many authors use terminology different from ours, some align partially with our usage. Schervish (1995), uses "Kullback-Leibler information" for I, which is more descriptive than the widely used "Kullback-Leibler divergence" Kullback & Leibler (1951), Bishop (2006). In particular, Kullback (1959) refers to I as an "information function." Donsker & Varadhan (1975) denote their rate function by I rather than H; the relative information is the i.i.d. special case of their I.

Doob's 1934 proof appeared at the dawn of modern probability theory, within the milieu of Daniell, Fréchet, Hopf, Khintchine, Kolmogorov, Lévy, and Wiener, and within a year of the Kolmogorov (1933) extension theorem.

5 Multivariate Normal

A symmetric matrix Q is *positive*, $Q > 0$, if $v \cdot Qv > 0$ for $v \neq 0$. A symmetric matrix is positive iff its eigenvalues are positive. The *determinant* $\det Q$ is the product of the eigenvalues of Q.

Let μ be in $\mathbf{R}^d$, Q a positive $d \times d$ matrix, $\theta = (\mu, Q)$, and let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With X in $\mathbf{R}^d$, the *normal density with mean μ and variance Q* is $p(X, \theta) = e^{-S(X,\theta)}$

where

$$S(X,\theta) = \frac{1}{2}\log\det(2\pi Q) + \frac{1}{2}(X-\mu)\cdot Q^{-1}(X-\mu). \tag{11}$$

This defines the parametric family.

A random variable X in $\mathbf{R}^d$ is *normally distributed with parameter* $\theta^* = (\mu^*, Q^*)$ if its moment-generating function is

$$E_*\left(e^{v\cdot X}\right) = \exp\left(v\cdot\mu^* + \frac{1}{2}v\cdot Q^* v\right). \tag{12}$$

Suppose X is normally distributed with parameter $\theta^* = (\mu^*, Q^*)$ as in (12), and let X_1, X_2, ... be independent and identically distributed copies[1] of X. We show θ^* is the true parameter, and we verify A, B, and C.

We say a symmetric matrix Q is *nonnegative*, and we write $Q \ge 0$, if $v\cdot Qv \ge 0$ for all v. Given symmetric matrices Q, R we write $Q \ge R$ if $Q - R \ge 0$. If $Q > 0$, then $Q \ge \delta I$ for some $\delta > 0$. The *trace* of a square matrix is the sum of its diagonal entries. When the matrix is symmetric, its trace is the sum of its eigenvalues. If Q and R are symmetric, $\text{trace}(QR) = \text{trace}(RQ)$. If $Q \ge 0$ and $R \ge S$, then $\text{trace}(QR) \ge \text{trace}(QS)$.

Let $\|Q\| = \sqrt{\text{trace}(Q^2)}$ be the *norm* of Q. Then $\|Q\|^2$ is the sum of the squares of the entries of Q and we have the *Cauchy-Schwarz inequality*

$$|\text{trace}(QR)| \le \|Q\|\,\|R\|, \qquad Q, R \text{ symmetric.}$$

In particular, $\text{trace}(IZ) \le \|I\|\,\|Z\|$. Since $\text{trace}(I) = d$ and $\text{trace}(Z^2)$ is the sum of the eigenvalues squared of Z,

$$\sum_{i=1}^{d} \lambda_i \le \sqrt{d(\lambda_1^2 + \lambda_2^2 + \cdots + \lambda_d^2)}. \tag{13}$$

Given vectors v, w in $\mathbf{R}^d$, the *tensor product* $v\otimes w$ is the matrix whose ij-th entry is $v_i w_j$. Then $v\otimes v$ is nonnegative and

$$R = v\otimes v \qquad \implies \qquad \text{trace}(QR) = v\cdot Qv.$$

Since $R^2 = |v|^2 R$ and $\text{trace}(R) = |v|^2$, $\|R\| = |v|^2$. Using this and Cauchy-Schwarz,

$$\delta = \|Q\| \qquad \implies \qquad Q \le \delta I. \tag{14}$$

[1] This setup is the i.i.d. special case of the Kolmogorov extension theorem.

If $u = \sqrt{\epsilon}v - w/\sqrt{\epsilon}$, then $u \otimes u \geq 0$, hence

$$v \otimes w + w \otimes v \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w. \tag{15}$$

Let $\theta' = (\mu', Q')$, $g(X) = 1 + |X|^2$ and let $\delta(\theta, \theta')$ equal half of

$$|\log \det Q - \log \det Q'| + \left|Q^{-1}\mu - (Q')^{-1}\mu'\right| + \left\|Q^{-1} - (Q')^{-1}\right\| + \left|\mu \cdot Q^{-1}\mu - \mu' \cdot (Q')^{-1}\mu'\right|.$$

Since[2] the determinant is continuous, $\delta(\theta, \theta')$ is continuous. Now δ consists of four terms. By the triangle inequality and Cauchy-Schwarz, $|S(X, \theta) - S(X, \theta')|$ is no larger than the sum of the first term, $2|X|$ times the second term, $|X|^2$ times the third term, and the fourth term. Since $2|X| \leq g(X)$, (8) follows from (11), establishing C.

The proof of the following is in the appendix.

Proposition. *For $Q^* > 0$ and c scalar, let $\epsilon = e^{-2c-1} \det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \text{trace}(Q^{-1}Q^*), \qquad Q > 0. \tag{16}$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\epsilon Q^* \leq Q \leq Q^*/\epsilon$.

Let $f(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu)$. Given $\theta = (\mu, Q)$, let

$$F(\mu, Q) = F_*(Q, f(\mu)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

By the proposition, (μ^*, Q^*) is the unique minimizer of $F(\mu, Q)$.

In (12), multiply both sides by $e^{-v \cdot \mu}$, then replace v by tv and differentiate twice at $t = 0$. This yields

$$E_*(X) = \mu^*, \qquad E_*\left((X - \mu) \otimes (X - \mu)\right) = f(\mu).$$

It follows $E_*(S(X, \theta)) = F(\mu, Q)$, and $\theta^* = (\mu^*, Q^*)$ is the true parameter, establishing A.

The second consequence of the proposition is properness of $F(\mu, Q)$. If $F(\mu, Q) \leq c$, then $F_*(Q, Q^*) \leq c$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$, confining Q to a compact subset of the space of positive definite matrices. From this,

$$\begin{aligned} c &\geq F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu) \\ &\geq F_*(Q^*, Q^*) + \frac{\epsilon}{2}(\mu^* - \mu) \cdot (Q^*)^{-1}(\mu^* - \mu), \end{aligned}$$

[2] See Lemma 3 in the appendix.

confining μ to a compact subset of $\mathbf{R}^d$. Thus $F(\mu, Q)$ is proper.

Now let $0 < \epsilon < 1/2$, and let

$$f_\epsilon(\mu) = (1-\epsilon)f(\mu) - \epsilon Q^*, \qquad F_\epsilon(\theta) = F_\epsilon(\mu, Q) = F_*(Q, f_\epsilon(\mu)).$$

Since $f_\epsilon(\mu) \geq (1-2\epsilon)f(\mu)$,

$$F_\epsilon(\mu, (1-2\epsilon)Q) \geq \frac{1}{2}\log\det((1-2\epsilon)I) + F(\mu, Q),$$

hence $F_\epsilon(\theta)$ is proper.

If Q_n is a sequence of symmetric matrices with $Q_n \to 0$, then $\|Q_n\| \to 0$. By (14), given $Q > 0$, there is $N \geq 1$ such that $Q_n \leq Q$ for $n \geq N$.

With

$$Q_n(\mu) = \frac{1}{n}\sum_{k=1}^n (X_k - \mu) \otimes (X_k - \mu), \qquad \hat{\mu}_n = \frac{1}{n}\sum_{k=1}^n X_k, \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$.

By the LLN, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \to 0, \qquad Q^* - Q_n(\mu^*) \to 0, \qquad \text{as } n \to \infty.$$

Thus, from above, there is $N \geq 1$, depending on the samples, such that, with probability one,

$$(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon^2 Q^*, \qquad Q_n(\mu^*) \geq (1-\epsilon)Q^*, \qquad \text{for } n \geq N. \tag{17}$$

Fix samples for which (17) holds. We establish (6) for these samples and $F(\theta) = F_\epsilon(\theta)$.

With $w = \hat{\mu}_n - \mu^*$ and $v = \mu - \mu^*$, using (15),

$$v \otimes w + w \otimes v \leq \epsilon(\mu^* - \mu) \otimes (\mu^* - \mu) + \frac{1}{\epsilon}(\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon f(\mu)$$

for all μ and $n \geq N$. Writing $X_k - \mu = (X_k - \mu^*) - v$,

$$Q_n(\mu) = Q_n(\mu^*) - (v \otimes w + w \otimes v) + v \otimes v.$$

Combined with the above inequalities, this leads to

$$Q_n(\mu) \geq f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \geq N.$$

In particular, since $f_\epsilon(\mu) \geq (1-2\epsilon)Q^*$, we have shown $\hat{Q}_n > 0$ for $n \geq N$.

Since $S_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \geq \hat{Q}_n$ by least squares, by the proposition,

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_*(Q, \hat{Q}_n) \geq F_*(\hat{Q}_n, \hat{Q}_n) = S_n(\hat{\theta}_n),$$

for all θ and $n \geq N$. Thus $\hat{\theta}_n$ is the MLE for $n \geq N$, and

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \geq F_*(Q, f_\epsilon(\mu)) = F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \geq N.$$

This establishes (6) and completes the derivation of B.

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A Auxiliary Proofs

Proof of (7) *with probability one.* By separability of the parameter space, there is a countable dense set D of parameters. Since D is countable, with probability one,

$$S_n(\theta') \to E_*(S(X, \theta')), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta' \text{ in } D. \tag{18}$$

Fix samples for which (9) and (18) hold. We establish (7) for these samples. If θ, θ' are any parameters, by (8) and the triangle inequality,

$$\begin{aligned} |S_n(\theta) - E_*(S(X,\theta))| \leq& |S_n(\theta') - E_*(S(X,\theta'))| \\ &+ \delta(\theta,\theta')\left(E_*(g(X)) + \frac{1}{n}\sum_{k=1}^{n} g(X_k)\right). \end{aligned} \tag{19}$$

Now take the $\limsup_{n\to\infty}$ in (19). By (9) and (18),

$$\limsup_{n\to\infty} |S_n(\theta) - E_*(S(X,\theta))| \leq 2\delta(\theta,\theta')E_*(g(X)), \quad \text{for all } \theta' \text{ in } D \text{ and all } \theta.$$

Since D is dense, $\delta(\theta,\theta')$ is continuous, and $\delta(\theta,\theta) = 0$, for fixed θ and varying θ', the right side is arbitrarily small, while the left side does not depend on θ'. This establishes (7). □

To derive the proposition, we need four Lemmas. The proofs below are included to keep the paper self-contained: the point is not merely to verify the required facts, but to expose the elementary mechanism behind them.

By the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v \otimes v$, and every positive matrix is a sum of δI and matrices of the form $v \otimes v$.

Lemma 1. *With Q invertible symmetric and t a scalar,*

$$\det(Q + tv \otimes v) = \det(Q)\left(1 + tv \cdot Q^{-1}v\right).$$

Proof. Assume $t \neq 0$. If v is orthogonal to an eigenvector x of Q, then we reduce to a lower-dimensional case by restricting to $x^\perp$. Hence we may assume $v \cdot x \neq 0$ for all eigenvectors x of Q. If v is orthogonal to an eigenvector x' of $Q' = Q + tv \otimes v$, then x' is an eigenvector of Q, contradicting our assumption. Hence we may assume $v \cdot x' \neq 0$ for all eigenvectors x' of Q'. If x'_1 and x'_2 are linearly independent eigenvectors of Q' corresponding to the same eigenvalue, then, for suitable r_1, r_2, $x' = r_1 x'_1 + r_2 x'_2$ is an eigenvector of Q' satisfying $v \cdot x' = 0$, contradicting our assumptions. Hence we may assume the eigenvalues of Q' are simple. If x and x' are eigenvectors for Q and Q' corresponding to the same eigenvalue, then

$$t(v \cdot x')(v \cdot x) = x' \cdot (tv \otimes v)x = x' \cdot Q'x - x' \cdot Qx = 0,$$

contradicting our assumptions. Hence we may assume the eigenvalues of Q and the eigenvalues of Q' do not overlap. Let $\lambda_1, \lambda_2, \dots, \lambda_d$ be the eigenvalues of Q and $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ the eigenvalues of Q'. Let $v_1, v_2, \dots, v_d$ be the corresponding orthonormal basis of eigenvectors of Q. Then the *secular equation* is

$$\sigma(z) = 1 + tv \cdot (Q - zI)^{-1}v = 1 + t\sum_{i=1}^{d} \frac{(v \cdot v_i)^2}{\lambda_i - z} = 0.$$

If λ' is an eigenvalue of Q' with eigenvector x', then $(Q' - \lambda' I)x' = 0$, hence $(Q - \lambda' I)x' + t(v \cdot x')v = 0$. Taking the dot product with $(Q - \lambda' I)^{-1}v$ yields $v \cdot x' + t(v \cdot x')(v \cdot (Q - \lambda' I)^{-1}v)$, hence $\sigma(\lambda') = 0$. If we let

$$\det(Q - zI) = \prod_{i=1}^{d}(\lambda_i - z), \qquad p(z) = \det(Q - zI)\sigma(z),$$

then $p(z)$ is a polynomial and $p(\lambda') = 0$. Since $\lambda'_1, \lambda'_2, \dots, \lambda'_d$ are simple, $p(z) = \prod_{i=1}^{d}(\lambda'_i - z)$. Since the product of the roots is $p(0)$, the identity follows. □

Lemma 2. *If $S > 0$, $Z > 0$, then* $\det(SZS) = \det(Z)\det(S)^2$.

Proof. Assume the identity holds for Z. By Lemma 1,

$$\begin{aligned}\det(S(Z + v \otimes v)S) &= \det(SZS + Sv \otimes Sv)\\ &= \det(SZS)(1 + Sv \cdot (SZS)^{-1}Sv)\\ &= \det(Z)\det(S)^2(1 + v \cdot Z^{-1}v) = \det(Z + v \otimes v)\det(S)^2,\end{aligned}$$

hence the identity holds for $Z + v \otimes v$ replacing Z. Also the identity holds for $Z = \delta I$. Since every $Z > 0$ is a sum of δI and a sum of matrices of the form $v \otimes v$, the identity holds for all $Z > 0$ by induction over the number of terms in the sum. □

Lemma 3. *The determinant* $\det Q$ *is a continuous function of* $Q > 0$.

Proof. Let Z be symmetric and $Q > 0$. The continuity of $\det Q$ at $Q = I$ follows from the inequality

$$1 + |\det(I + Z) - 1| \le \exp\left(\sqrt{d}\,\|Z\|\right). \tag{20}$$

Indeed, if $Z = Z_n \to 0$, then $\det(I + Z_n) \to 1 = \det I$, establishing continuity at $Q = I$. For continuity at Q, write $Q = S^2$ with $S > 0$ and set $W = S^{-1}ZS^{-1}$. Then by Lemma 2, if $I + W > 0$,

$$|\det(Q + Z) - \det Q| = \det Q \cdot |\det(I + W) - 1|.$$

If $Z_n \to 0$, then $W_n = S^{-1}Z_nS^{-1} \to 0$, hence $I + W_n > 0$ eventually and $\det(Q + Z_n) \to \det Q$, establishing continuity at Q. It remains to establish (20).

If λ_1, λ_2, ..., λ_d are the eigenvalues of Z, then

$$\det(I + Z) = \prod_{i=1}^{d}(1 + \lambda_i), \qquad \operatorname{trace}(Z^2) = \sum_{i=1}^{d}\lambda_i^2, \qquad \|Z\| = \sqrt{\sum_{i=1}^{d}\lambda_i^2},$$

hence (20) reduces to

$$1 + \left|\prod_{i=1}^{d}(1 + \lambda_i) - 1\right| \le \exp\left(\sqrt{d(\lambda_1^2 + \lambda_2^2 + \cdots + \lambda_d^2)}\right). \tag{21}$$

To establish (21), let $s(\lambda_1, \lambda_2, \ldots, \lambda_d)$ be the sum of subproducts of λ_1, λ_2, ..., λ_d obtained by expanding $\prod_{i=1}^{d}(1 + \lambda_i)$. Then $s(\lambda_1, \lambda_2, \ldots, \lambda_d)$ contains exactly one constant term 1. Subtracting 1, then applying the triangle inequality, then adding 1 back, the left side of

(21) is no larger than $s(|\lambda_1|, |\lambda_2|, \ldots, |\lambda_d|)$, which equals $\prod_{i=1}^d (1+|\lambda_i|)$. Since $1 + t \leq e^t$ for $t \geq 0$, the left side of (21) is no larger than

$$\prod_{i=1}^d (1+|\lambda_i|) \leq \prod_{i=1}^d e^{|\lambda_i|} = \exp\left(\sum_{i=1}^d |\lambda_i|\right) \leq \exp\left(\sqrt{d(\lambda_1^2 + \lambda_2^2 + \cdots + \lambda_d^2)}\right),$$

where we used (13) with $|\lambda_i|$ replacing λ_i. This establishes (21), hence (20). □

Lemma 4. *For c scalar, let $\epsilon = e^{-c}$ and $h(\lambda) = \log\lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$, and $h(\lambda) \leq c$ implies $\epsilon \leq \lambda \leq 1/\epsilon$.*

Proof. The first part is checked by differentiation. For the second part, clearly $h(\lambda) \leq c$ implies $\lambda \leq 1/\epsilon$. Since $\lambda \log\lambda \geq -1/e$ and $(e-1)e^c \geq ec$ for $c \geq 1$, $h(\lambda) \leq c$ and $c \geq 1$ implies $\lambda \geq (1-1/e)/c \geq \epsilon$. □

Proof of the proposition. Since $Q^* > 0$, there is $S > 0$ with $Q^* = S^2$. Let $Z = S^{-1}QS^{-1}$ and let λ_1, λ_2, ..., λ_d be the eigenvalues of Z. Then $\operatorname{trace}(Q^{-1}Q^*) = \operatorname{trace}(SQ^{-1}S) = \operatorname{trace}(Z^{-1})$. By Lemmas 2 and 4,

$$\begin{aligned} 2F_*(Q,Q^*) - 2F_*(Q^*,Q^*) &= \log\det Q - \log\det Q^* + \operatorname{trace}(Q^{-1}Q^* - I) \\ &= \log\det(SZS) - \log\det(S)^2 + \operatorname{trace}(Z^{-1} - I) \\ &= \log\det Z + \operatorname{trace}(Z^{-1} - I) \\ &= \sum_{i=1}^d (h(\lambda_i) - 1) \geq 0, \end{aligned} \tag{22}$$

establishing $F_*(Q,Q^*) \geq F_*(Q^*,Q^*)$. If (22) is an equality, $Z = I$ hence $Q = Q^*$. Now suppose $F_*(Q,Q^*) \leq c$. Then

$$2F_*(Q,Q^*) - 2F_*(Q^*,Q^*) \leq 2c - \log\det(2\pi Q^*) - \operatorname{trace}(I) = \log(1/\epsilon) - 1.$$

Since the terms in the sum in (22) are nonnegative, $h(\lambda_i) - 1 \leq \log(1/\epsilon) - 1$ for $i = 1, 2, \ldots, d$. By Lemma 4, $\epsilon \leq \lambda_i \leq 1/\epsilon$ for $i = 1, 2, \ldots, d$. This implies $\epsilon I \leq Z \leq I/\epsilon$, hence $\epsilon Q^* \leq Q \leq Q^*/\epsilon$. □

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